

Age-Related Changes in Human Left and Right Atrial Conduction

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Age-Related Changes in Atrial Electrophysiology. *Introduction:* Advancing age is an independent risk factor for atrial fibrillation (AF), which is considered to be initiated by ectopic triggers and maintained by an arrhythmogenic substrate. It is not known whether substrate changes produce this age-related increase in propensity toward AF. We addressed the hypothesis that advancing age is associated with changes in biatrial electrophysiology even in patients with no history of atrial arrhythmias.

Methods and Results: Patients with left-sided accessory pathways and requiring routine electrophysiological studies were recruited. Electroanatomic mapping was performed in the left and right atria of 23 patients (age ranging from 17 to 75 years) with structurally normal hearts and no history of AF during sinus rhythm and pacing. Unlike previous studies, a trigonometric method was used to quantify wavefront propagation velocities (WPV) precisely in the direction of propagation. Refractoriness was measured at 2 cycle lengths, at three different atrial sites. Both right ($r = -0.77$, $P < 0.0001$) and left ($r = -0.79$, $P < 0.001$) atrial WPV demonstrated strongly inverse correlation with age. Furthermore, left and right WPVs were highly correlated ($r = 0.66$, $P < 0.01$), with velocities being 6.4 ± 2.2 cm/sec higher in the right atria ($P < 0.01$). Refractoriness was significantly correlated with increasing age only at the septum ($r = 0.53$, $P < 0.01$). Left atrial wavelength was inversely correlated with increasing age ($r = -0.56$, $P = 0.03$). P wave duration was associated with age ($r = 0.42$, $P = 0.04$) and left atrial size ($r = 0.44$, $P = 0.04$) but not atrial WPV.

Conclusion: Aging human atria demonstrate progressive decline in WPV and increase in septal refractoriness. These age-related changes in biatrial electrophysiology are likely to be important factors in the age-related increase in AF prevalence. (*J Cardiovasc Electrophysiol*, Vol. 17, pp. 120-127, February 2006)

aging, atrium, conduction, electrophysiology, mapping

Introduction

Advancing age is an important independent risk factor for atrial fibrillation (AF). Population-based studies have shown that for each decade of advancing age, the risk of developing AF, including postoperative AF more than doubles.^{1,2} The pathological and functional changes in the atria that lead to this age-related predisposition to AF are not well understood. Although current theories of AF suggest that left atrial triggers are important in the initiation of human AF, it is not known whether the increasing prevalence with age is related to left atrial triggers or the development of an arrhythmogenic substrate.³

Classical models of arrhythmogenesis predict that shortening of refractoriness and slowing of conduction are proarrhythmic. Although patients with AF accordingly demonstrate shortening of refractory periods, most previous studies

of normal subjects have concluded that atrial refractoriness increases with age, which would be expected to protect from AF.⁴⁻⁹ Changes in atrial conduction velocity (CV) with age might therefore be expected to be proarrhythmic to offset the effects of prolonged refractoriness.

Although conduction is stochastic at the microscopic level, in the intact atrium this microscopic irregularity is smoothed out into uniform wavefronts. Wavefront propagation velocity (WPV) can only be measured accurately if the direction of the wavefront and the time taken to travel a specific distance in that direction can be determined. Previous studies measuring WPV by dividing the distance by the activation interval between two points not necessarily in line with the direction of wavefront propagation would overestimate the true velocity and introduce artefactual variation in the data. Accordingly, measurements of WPV from previous studies with endocardial mapping techniques have tended to show exaggerated CVs (up to twofold) and greater variances (up to threefold), compared with studies using simultaneous multi-electrode intraoperative epicardial mapping in which the direction of wavefront propagation can be taken into account but which are subject to the limitations of studying limited areas of atrial free wall in the anesthetized patient.^{6,10-15} Unlike previous studies investigating endocardial CV, we validated an algorithm that incorporated the local vector of activation to measure WPV in the direction of wavefront propagation from three-dimensional electroanatomical maps to test the hypothesis that there are age-related alterations in WPV in both atria.

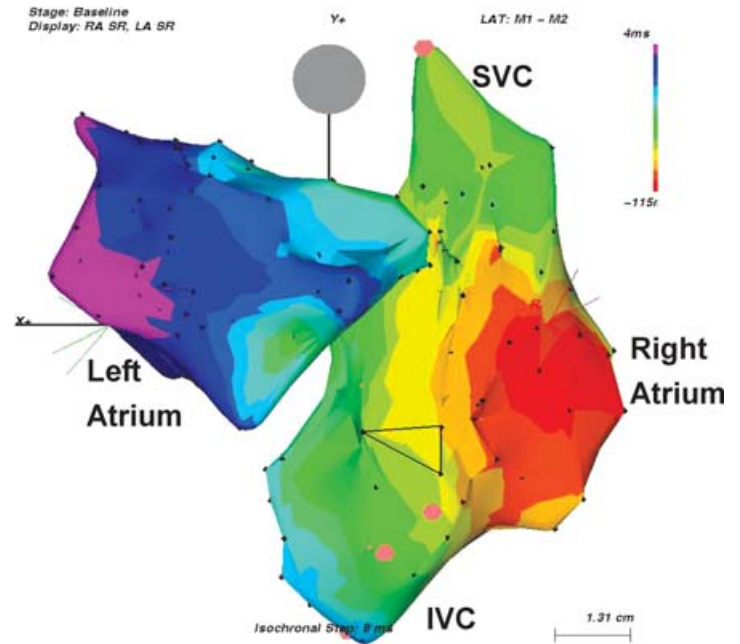
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Figure 1. Carto™ isochronal maps of bia-atrial activation during sinus rhythm in the PA projection, with earliest activation in the lateral right atrium (in red) and latest at the lateral mitral annulus (in purple). Most points were collected along the medial, lateral, and posterior aspects of the right atria and the roof and posterior wall of the left atria. Three-dimensional coordinates and local activation times of the three points bounded by the triangle were used to calculate the local wavefront propagation velocity. SVC = superior vena cava; IVC = inferior vena cava.



Methods

Patients undergoing clinically indicated that electrophysiological studies (EPS) were recruited. Patients were excluded if they had an abnormal echocardiogram, impaired left ventricular function, history of AF, valvular heart disease, and ischemic heart disease or at any stage received amiodarone therapy. Antiarrhythmic drugs were ceased at least 5 half-lives before EPS. All patients gave written and informed consent at least 2 weeks before the study, which was granted institutional ethics approval.

All patients had 72-hour Holter recordings prior to EPS and during follow-up to exclude atrial tachycardias. Other cardiac investigations such as coronary angiograms were performed if clinically indicated.

Electrophysiological Study

EPS was undertaken in the postabsorptive state and all measurements were made prior to the onset of ablation. A decapolar catheter (Bard Woven, Lowell, MA, USA) with 2-mm interelectrode spacing was positioned in a stable position within the coronary sinus (CS) to provide reference electrograms. Left atrial mapping for activation analysis and ablation was performed via a transseptal sheath if left atrial access was clinically indicated.

Surface electrocardiogram (ECG) and bipolar endocardial electrograms were monitored continuously and stored on a digital amplifier/recorder system (Siemens EP, Munich, Germany) for offline analysis. Intracardiac bipolar electrograms from conventional electrophysiological catheters were filtered from 30 to 500 Hz. Ten consecutive P wave durations of lead II at the start of the procedure were measured with computer-assisted calipers at a display speed of 200 mm/sec.

Electroanatomic Mapping

The Carto™ system (Biosense Webster, Inc., Diamond Bar, CA, USA) has been described in detail before.¹⁶ Briefly, global activation mapping was recorded during a stable

rhythm by point-to-point acquisition of positions with reliable local activation times and accurate three-dimensional coordinates, gated to a stable reference CS electrogram. The maximum deflection from baseline of bipolar electrograms of each point defined the local activation time (LAT). If the bipolar electrograms have equivalent maximum positive and negative deflections, the maximum negative $\delta V/\delta t$ of the unipolar electrogram was used to mark the LAT.¹⁷ Signals from the Carto Navistar™ mapping catheter (Biosense Webster) were filtered between 1 Hz and 240 Hz for unipolar electrograms and between 30 Hz and 400 Hz for bipolar electrograms. Each point was acquired with the auto-freeze mode only if end-diastolic stability was less than 2 mm and the LAT stability was <2 msec between two consecutive measurements, implying a stable catheter-wall contact. Every point was scrutinized during offline analysis to ensure that the LAT were correctly annotated according to the criteria described above.

Isochronal activation maps were created for both atria during sinus rhythm (SR) and in the left atrium during distal CS pacing at fixed cycle length (CL) of 600 msec (the longest CL that would enable consistent atrial capture in all patients) at twice diastolic threshold (Fig. 1). Mapping the left atrium during distal CS pacing was used to avoid the influence of varying left atrial breakthrough from right atrial pacing.¹⁸

If atrioventricular reentrant tachycardia was initiated during the protocol, termination was achieved by sensed ventricular extrastimuli, without pharmacological or electrical cardioversion. Sinus rhythm cycle length (SR CL) was measured as the average of all preceding beat-to-beat CL preceding each point acquired during mapping.

Calculation of Wavefront Propagation Velocity

The principle of triangulation was applied to calculate the WPV of a wavefront in the direction of propagation.¹¹ Adjacent points (less than 20 mm apart) that were activated by the same wavefront but separated by a LAT of more than 3 msec apart were chosen from a large collection of points in each chamber and grouped into triads.

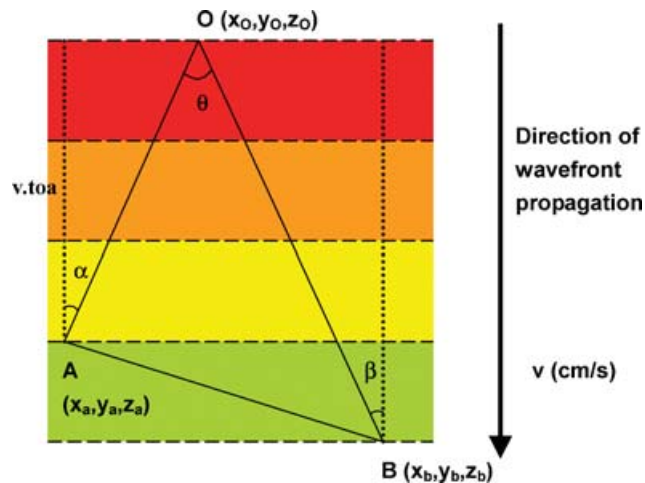


Figure 2. Schematic representation of a triad of three endocardial points, marked as O, A, and B. v represents wavefront propagation velocity.

A triad of points would define a triangle in a single plane (Fig. 2). The difference in LATs and the distances between these points were used to measure the WPV (see Appendix for full details).

Wavefront propagation velocity, $v = \frac{OA}{tOA} \cos \alpha$ (Fig. 2).

The distance between any two points in three-dimensional space could be calculated using the formula,

$OA = \text{distance between O and A} = \sqrt{(x_a - x_o)^2 + (y_a - y_o)^2 + (z_a - z_o)^2}$, if O and A had the coordinates (x_o, y_o, z_o) and (x_a, y_a, z_a) , respectively.

tOA was the difference in LAT between O and A and α was calculated by the formulae described in the Appendix. These formulae were programmed into an Excel macro to calculate WPV. In this model, conduction within each triad was assumed to be uniform and chamber curvature was not taken into account, but by limiting the distance between two furthest points of the triad to less than 20 mm, with a mean interpoint distance of 8.6 ± 5.1 mm, these potential forces of inaccuracy were minimized.

In order to validate both the algorithm and location software, a cardboard strip was fashioned and parallel lines with 5-mm spacing were drawn to simulate a hypothetical wavefront traveling at 100 cm/sec. The Navistar catheter and reference patches were used after the clinical procedure in 5 patients to measure the three-dimensional coordinates of any three random points positioned along different lines. These data together with the relative "activation times" of these points (i.e., the time taken by the wavefront to travel from one line to the next was arbitrarily set at 5 msec) were entered into the triangulation algorithm to determine the calculated WPV. This process was repeated ten times with each catheter and the estimated WPV was 100.1 ± 1.6 cm/sec.

Measurement of Refractoriness

Effective refractory periods (ERP) of the atria were assessed at three separate sites (lateral high right atrium, right atrial septum, and distal CS) at two different CLs of 600 msec and 500 msec, at twice diastolic thresholds and pulse width of 2.0 msec with an eight-beat drive train followed by a S_2 extrastimulus. ERP was defined as the longest S_1 – S_2 coupling interval that resulted in failure of local capture.^{6,7} The

adaptation of refractoriness was calculated as the difference between ERP at the two different CLs at each site. Dispersion of refractoriness was the maximum difference of ERP between the three sites at each CL.

Wavelength

The wavelength of each atrium was calculated as the product of the ERP and WPV at the same CL. Right atrial conduction was measured only during SR and therefore right atrial wavelength was approximated by multiplying the right atrial WPV and ERP measured at lateral right atrium during pacing at 600 msec.

Statistical Analysis

Correlation was analyzed by Pearson's correlation tests. Comparison between paired values was performed with paired t -tests. Figures are expressed as mean $\pm$ 1 standard deviation. Statistical significance was established at $P < 0.05$.

Results

The protocol was completed in 23 patients. Eighteen had left-sided accessory pathways (15 overt, 3 concealed) and 1 had a right-sided accessory pathway. In these patients, EPS and ablation were indicated due to occurrence of symptomatic atrioventricular reentrant tachycardias. Four remaining patients had EPS for investigation of syncope or palpitations without documented tachyarrhythmias, and all had normal EPS with no inducible tachycardias (Table 1). The mean age was 47 years (range: 17–75 years) and 13 were male. All patients had a structurally normal heart as defined by echocardiography with no episodes of AF detected during pre- and postprocedural Holter monitoring.

Wavefront Propagation Velocity

In total, 23 right atria and 15 left atria were mapped with an average of 92 ± 25 points per chamber. The mean left and right atrial WPV were 70.2 ± 9.9 and 77.0 ± 10.8 cm/sec, respectively (Table 2). Age was negatively correlated with WPV in both atria during SR (RA $r = -0.77$, $P < 0.0001$, LA $r = -0.79$, $P < 0.001$) and in the left atrium during pacing ($r = -0.65$, $P < 0.01$) (Fig. 3). SR CL did not correlate with either patients' age ($r = 0.29$, $P = \text{NS}$) or WPV (RA $r = -0.30$, LA $r = 0.29$; $P = \text{NS}$). WPV of the two atria during SR was significantly correlated ($r = 0.66$, $P < 0.01$), although mean right atrial WPV was 6.4 ± 2.2 cm/sec greater than paired left atrial measurements ($n = 15$, $P < 0.01$), with no significant change in SR CL (RA -801 ± 134 msec vs LA -792 ± 138 msec, $P = \text{NS}$) during mapping (Fig. 4).

Atrial Refractoriness

ERPs showed a significant correlation with age at both CLs when measured at the atrial septum ($r = 0.53$, $P < 0.01$ at 600 msec; $r = 0.48$, $P = 0.02$ at 500 msec) (Fig. 5). There was no correlation with age at neither the high right atrium ($r = 0.22$ at 600 msec, $r = 0.18$ at 500 msec; $P = \text{NS}$), and left atrium ($r = 0.19$ at 600 msec, $r = 0.16$ at 500 msec; $P = \text{NS}$) nor did dispersion ($r = 0.27$ at 600 msec, $r = 0.12$ at 500 msec; $P = \text{NS}$) and rate-adaptation of refractoriness ($r = 0.22$ at high right atrium, $r = 0.39$ at septum, $r = 0.16$ at distal CS; $P = \text{NS}$) correlate with age.

TABLE 1
Patient Characteristics

Patient	Age (Years)	Sex	LVED (mm)	LA (mm)	LVEF (%)	Indication for EPS
1	17	M	41.5	31.1	80.9	WPW (LAP)
2	24	F	40.6	30.8	68.1	Concealed LAP
3	29	F	38.8	30.2	84.3	WPW (LAP)
4	33	M	47.7	39.6	81.1	WPW (LAP)
5	34	F	41.1	35.3	55	Syncope
6	34	F	38.5	29.6	70.2	WPW (LAP)
7	35	M	56.2	35.6	71.5	WPW (LAP)
8	36	M	42.5	32.0	71.5	WPW (LAP)
9	42	F	43.0	36.0	82.4	WPW (RAP)
10	43	F	48.3	34.5	79.1	Palpitations
11	44	M	48.1	30.7	72.5	WPW (LAP)
12	47	F	41.0	34.2	78.4	Palpitations
13	49	M	45.5	39.0	75.7	WPW (LAP)
14	52	F	36.7	39.2	72.0	WPW (LAP)
15	55	M	48.4	36.0	66.0	Syncope
16	56	M	40.8	29.2	74.5	WPW (LAP)
17	56	M	57.7	39.6	81.1	WPW (LAP)
18	56	M	32.3	33	66.3	WPW (LAP)
19	63	M	51.4	39.0	67.7	WPW (LAP)
20	67	M	44.2	37.4	73.2	Concealed LAP
21	67	F	32.0	40	73.2	Concealed LAP
22	71	F	54.0	41.2	64.8	WPW (LAP)
23	75	M	54.0	38.9	71.3	WPW (LAP)

EPS = electrophysiological studies; LA = left atrial size; LVED = left ventricular end-diastolic dimension; LVEF = left ventricular ejection fraction; WPW = Wolff-Parkinson-White syndrome; LAP = left-sided accessory pathway; RAP = right-sided accessory pathway.

Atrial Wavelength

Wavelength was inversely correlated with age in the left atrium ($r = -0.56$, $P = 0.03$) with a trend toward significance in the right atrium ($r = -0.39$, $P = 0.07$) (Fig. 6).

Left Atrial Dimension

Maximal anterior-posterior left atrial size (measured in the standard parasternal long-axis view) increased with advancing age ($r = 0.62$, $P < 0.01$).

TABLE 2
Electrophysiological Parameters*

Age (Years)	SR CL (msec)	Wavefront Propagation Velocity (cm/sec)			PWD (msec)	AERP at 600 msec (msec)			AERP at 500 msec (msec)		
		RA SR	LA SR	LA Paced		LRA	Septum	DCS	LRA	Septum	DCS
17	827	95.3	—	—	84	170	180	180	150	180	170
24	675	88.9	80.6	99.2	54	230	200	210	220	180	200
29	841	84.2	82.6	79.4	45	260	230	240	250	230	230
33	1,042	79.0	87.2	82.8	101	210	210	220	200	200	200
34	808	71.0	—	—	104	270	270	280	260	260	270
34	702	95.2	—	—	78	200	170	180	200	170	190
35	856	84.3	71.7	85.6	88	280	280	280	260	270	280
36	923	87.5	82.5	78.4	112	210	260	310	200	250	280
42	870	87.6	—	—	62	260	250	260	240	240	240
43	766	71.9	—	—	74	190	210	250	210	210	270
44	820	87.3	70.4	57.6	112	270	260	240	250	240	240
47	849	70.0	—	—	64	300	300	290	280	290	280
49	718	75.4	60.1	58.1	105	240	200	220	240	200	220
52	767	67.0	62.9	59.1	117	230	230	230	230	230	230
55	860	77.4	—	—	85	250	270	260	240	250	240
56	644	82.1	68.4	76.7	97	290	280	260	270	270	250
56	660	71.3	75.4	75.0	95	240	230	210	230	210	220
56	768	70.6	61.7	69.7	78	200	250	220	190	230	230
63	970	65.0	65.0	63.5	69	270	280	240	250	260	240
67	879	66.1	69.0	67.6	113	280	250	240	250	250	220
67	830	76.0	64.6	70.1	107	180	240	240	170	220	230
71	1,037	52.6	—	—	113	260	310	270	230	280	240
75	792	65.4	51.3	64.2	102	230	260	240	220	250	220

*Only mean values shown.

SR CL = sinus rhythm cycle length; RA = right atrium; LA = left atrium; SR = sinus rhythm; AERP = atrial effective refractory period; LRA = lateral right atrium; DCS = distal coronary sinus; PWD = P wave duration.

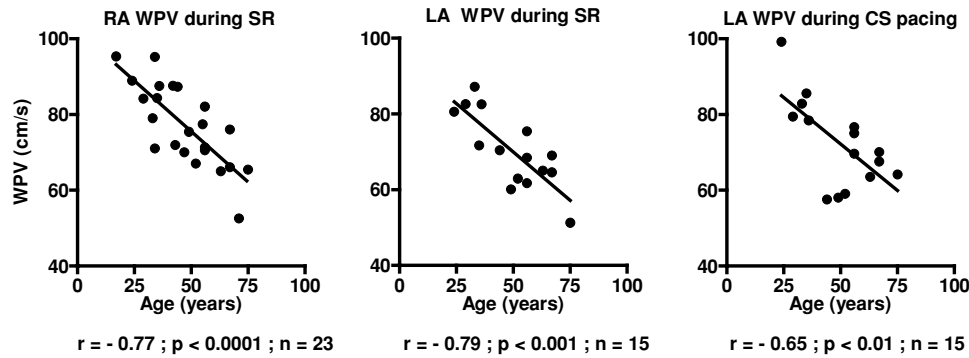


Figure 3. Correlation between atrial wavefront propagation velocities and age. WPV = wavefront propagation velocity; CS = coronary sinus; LA = left atrial; RA = right atrial; SR = sinus rhythm.

P Wave Duration

P wave duration in lead II showed a correlation with increasing age ($r = 0.42$, $P = 0.04$) and left atrial size ($r = 0.44$, $P = 0.04$). It, however, did not correlate with either left ($r = -0.31$, $P = \text{NS}$) or right atrial WPV ($r = -0.34$, $P = \text{NS}$) during SR.

Clinical Outcome

All patients remained free from AF for 12 ± 6 months following successful ablation in all cases.

Discussion

The current study demonstrates for the first time that WPV in both atria declines with increasing age. Furthermore, left atrial conduction is slower in comparison to the right atria. ERP was associated with increasing age only when measured at the septum. Left atrial wavelength decreases with advancing age.

Wavefront Propagation Velocity

Although catheter-based three-dimensional location systems have enabled accurate location of electrodes providing the basis for different methods for calculating CV, unless the direction of wavefront propagation is taken in consideration, assumed CV measurements will always tend to overestimate the true velocity and dispersion of conduction.^{6,10-15} These studies have found atrial CVs from 48 cm/sec up to 250 cm/sec compared with the much narrower range of 60–115 cm/sec with epicardial mapping, albeit limited by the inability to access all parts of the atria in anaesthetized open-chested humans. Uniquely, the technique described in the present study combines the advantage of being able to accurately measure WPV in the direction of wavefront propagation with the access afforded by endocardial mapping in the conscious patient, and was applied to measure WPV in the left atria as well as the right atria for the first time.

Age-related reduction of right atrial CV assessed only during distal CS pacing has recently been reported.⁶ We now report a similar pattern of age-related decline in left atrial WPV during pacing and in both atria during SR, independently of SR CL. Further analysis also revealed slower conduction in the left atria in comparison to the right atria.

The mechanism of age-related slowing of atrial conduction is likely to be multifactorial. Increased atrial fibrosis occurs in senescent animal atria, resulting in interatrial conduction delay and slowed conduction of premature beats.^{19,20} In isolated canine and human atrial fibers, the resultant separation of unit bundles by interstitial tissue leads to electrical uncoupling of side-to-side connections and electrical wavefronts propagating with greater complexity and increased anisotropy.^{21,22} Another structural disturbance associated with aging human hearts is amyloidosis that can reach a prevalence of more than 90% in the ninth decade.²³ Like fibrosis, the interstitial deposition of extracellular proteins could perturb conduction and has been associated with persistent AF in patients undergoing mitral valve surgery.²⁴ Alteration of autonomic function may also play a role with both cardiac sympathetic, assessed by positron emission tomography, and parasympathetic tone declining with age.^{25,26} While ex vivo tissue experiments have consistently shown greater average longitudinal but lower transverse conduction in older subjects, this study demonstrates for the first time that, in intact human hearts, biatrial WPV declines with age.^{21,22} It is plausible that atrial conduction slowing and macroscopic

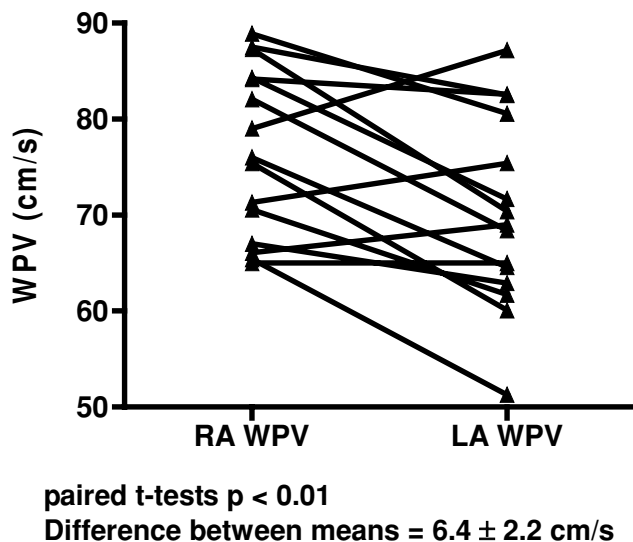


Figure 4. Comparison of paired left and right atrial wavefront propagation velocities during sinus rhythm. LA = left atrial; RA = right atrial; WPV = wavefront propagation velocity.

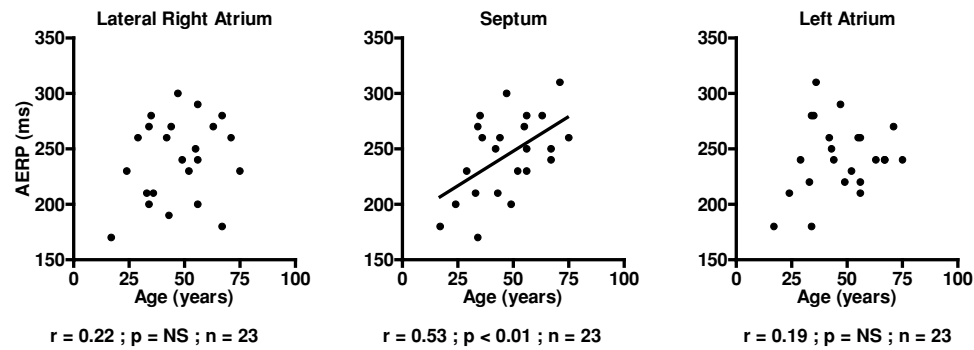


Figure 5. Relationship between atrial refractoriness at 600 msec and age. AERP = atrial effective refractory period.

structural changes, namely, left atrial enlargement may reflect the same pathological process.

The physiology underlying differences in WPV between different atrial chambers is also poorly understood. The pattern of atrial natriuretic peptide deposition and connexin expression differs between the left and right atria but the functional significance of such findings remains undetermined.^{27,28}

Refractoriness

In keeping with previous investigators, the degree of correlation between refractoriness and age was not uniform between different atrial sites.^{6,7} When ERP was measured at three sites along the lateral wall of the right atria in 17 normal subjects, ERP showed a direct correlation with age at only one of three sites.⁷ Although there is one report correlating age with dispersion of atrial refractoriness along the lateral wall of the right atria, our study have failed to demonstrate this association or any with adaptation of refractoriness.⁷

Atrial Wavelength

In order to interpret concomitant WPV and ERP changes, wavelength (the product of CV and ERP) has been proven to be a strong predictor of atrial arrhythmias, with wavelength shortening being considered to be proarrhythmic.²⁹ In our study, there was a decrease in left atrial wavelength with advancing age. Previous investigators have described age-related increases in refractoriness, which would be expected to afford protection against the proarrhythmic effects of conduction slowing. Indeed, in a large study of patients with no history of AF, induced AF occurred more commonly in younger patients and the greater degree of refractoriness in patients older than 70 years appeared to be a protective factor

against AF inducibility at EPS.⁴ Therefore, the overall decline of atrial wavelength with aging suggests that the degree of age-related proarrhythmic conduction slowing outweighs any potential antiarrhythmic effects afforded by prolongation of refractoriness. The additive effects of wavelength shortening and age-related left atrial enlargement would facilitate the maintenance of fibrillatory conduction, characteristic of AF.^{12,30}

Finally, prolongation of P wave duration has been suggested as evidence of proarrhythmic remodeling on the basis of intraatrial conduction slowing.³¹ Our observation that P wave duration did not correlate with atrial WPV, but was significantly associated with age and left atrial size, is in keeping with other reports that changes in P wave duration are more reflective of atrial volumes and interatrial conduction or dysynchrony rather than intraatrial conduction slowing.^{30,32,33}

Study Limitations

In order to study the electrophysiology of the left atria, generally considered the seat of the pathophysiology causing AF, we selected patients with left-sided accessory pathways enabling access to the left atrium.³ Patients were scrutinized for AF by reviewing all previous ECGs and 72-hour-Holter recordings prior to their study and during follow-up. All of our patients were free of AF during the procedure despite programmed stimulation and during clinical follow-up. It is possible that the atrial electrophysiology of patients with accessory pathways may not reflect the normal aging process but access to the left atrial would not have been ethically possible in a truly normal cohort, and although there have been no invasive studies comparing parameters of conduction between WPW patients without AF and age-matched normal subjects, P wave durations of the signal-averaged ECG have been shown to be similar in these two groups; although for reasons discussed above, P wave duration is at best a surrogate marker for interatrial conduction.³⁴

Conclusion

WPVs can be determined from three-dimensional activation maps using a novel algorithm that was designed to be used with endocardial mapping systems. Aging is characterized by slowing of conduction in both atria and increasing septal refractoriness. Such slowing of conduction results in proarrhythmic shortening of the atrial wavelength and is likely to contribute to the increased age-related prevalence of

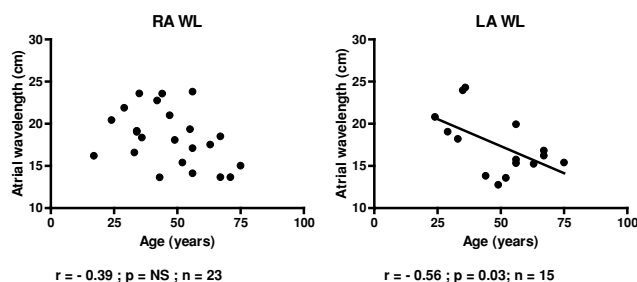


Figure 6. Relationship between atrial wavelength and age. LA = left atrial; RA = right atrial; WL = wavelength.

AF. Establishing the normal electrophysiological profile of aging may allow the identification of patients vulnerable to AF.

Appendix

Glossary (refer to Fig. 2)

OA = distance between O and A,

OB = distance between O and B,

AB = distance between A and B,

tOA = difference in LAT between O and A,

tOB = difference in LAT between O and B,

tAB = difference in LAT between A and B,

θ = angle subtended by OA and OB,

α = angle subtended by OA and the direction of wavefront propagation,

β = angle subtended by OB and the direction of wavefront propagation,

v = wavefront propagation velocity in the direction of wavefront propagation.

$$\cos \alpha = \frac{v \cdot tOA}{OA},$$

$$v = \frac{OA}{tOA} \cos \alpha. \quad (1)$$

Similarly,

$$\frac{v \cdot tOB}{OB} = \cos \beta = \cos(\theta - \alpha),$$

$$\frac{v \cdot tOB}{OB} = \cos \theta \cos \alpha + \sin \theta \sin \alpha,$$

$$v = \frac{OB}{tOB} (\cos \theta \cos \alpha + \sin \theta \sin \alpha). \quad (2)$$

By combining Equations 1 and 2,

$$\frac{OB}{tOB} (\cos \theta \cos \alpha + \sin \theta \sin \alpha) = \frac{OA}{tOA} \cos \alpha,$$

$$\cos \alpha = \frac{tOA \cdot OB}{tOB \cdot OA} (\cos \theta \cos \alpha + \sin \theta \sin \alpha),$$

$$\cos \alpha = \frac{tOA \cdot OB}{tOB \cdot OA} \cos \theta \cos \alpha + \frac{tOA \cdot OB}{tOB \cdot OA} \sin \theta \sin \alpha,$$

$$\cos \alpha \left(1 - \frac{tOA \cdot OB}{tOB \cdot OA} \cos \theta \right) = \sin \alpha \left(\frac{tOA \cdot OB}{tOB \cdot OA} \sin \theta \right),$$

$$\tan \alpha = \frac{\left(1 - \frac{tOA \cdot OB}{tOB \cdot OA} \cos \theta \right)}{\frac{tOA \cdot OB}{tOB \cdot OA} \sin \theta} = \frac{\frac{tOB \cdot OA}{tOA \cdot OB} - \cos \theta}{\sin \theta}.$$

For any straight line distance between two points in three-dimensional space, if O and A have the coordinates $(x_o, y_o,$

$z_o)$ and (x_a, y_a, z_a) , then distance between O and A = $OA = \sqrt{(x_a - x_o)^2 + (y_a - y_o)^2 + (z_a - z_o)^2}$.

Similarly, OB and AB can be calculated.

Using cosine rule, the value of θ is derived:

$$(AB)^2 = (OA)^2 + (OB)^2 - 2 \cdot \cos \theta (OA)(OB),$$

$$\cos \theta = \frac{(OA)^2 + (OB)^2 - (AB)^2}{2(OA)(OB)}.$$

Finally, v is derived by inserting the calculated values of α , tOA , and OA into Equation 1.

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